

URUP, Russian Federation

WMO: 321860

Lat: 46.200N

Lon: 150.500E

Elev: 76

StdP: 100.42

Time Zone: 10.00 (E10)

Period: 86-97

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-11.0	-9.8	-15.5	1.0	-9.7	-14.0	1.1	-9.0	24.1	-3.0	22.3	-3.8	9.6	320	0.957

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.0	17.2	14.9	15.5	14.0	14.1	12.9	15.8	16.4	14.4	15.1	13.2	13.9	5.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.5	11.1	16.2	14.1	10.1	14.7	12.9	9.3	13.4	44.7	16.4	40.7	15.2	37.5	13.9	22.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 20.5	17.8	15.6														
(5)			DB	-11.4	20.7	2.0	1.7	-12.8	21.9	-14.0	22.8	-15.1	23.8	-16.5	25.0	
			WB	-12.2	18.6	1.8	1.8	-13.5	19.9	-14.5	21.0	-15.5	22.0	-16.9	23.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	2.6	-4.3	-5.2	-3.1	0.2	3.0	5.0	7.9	11.1	9.8	6.8	1.7	-1.7	(6)
(7)		DBStd	5.83	2.78	2.94	2.16	1.87	2.08	1.97	2.20	2.64	2.06	2.07	2.70	2.52	(7)
(8)		HDD10.0	2769	444	426	406	294	217	150	74	19	27	101	251	363	(8)
(9)		HDD18.3	5726	702	659	664	544	475	399	325	224	254	356	501	621	(9)
(10)		CDD10.0	86	0	0	0	0	0	1	7	53	22	3	0	0	(10)
(11)		CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)
(12)		CDH23.3	1	0	0	0	0	0	0	0	1	0	0	0	0	(12)
(13)		CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	7.3	9.6	9.2	8.9	8.2	6.9	5.3	5.1	5.3	5.6	6.8	8.0	8.9	(14)
(15)	Precipitation	PrecAvg	1365	93	82	95	100	97	87	112	131	157	148	143	121	(15)
(16)		PrecMax	1706	144	204	178	181	148	156	225	306	254	234	296	191	(16)
(17)		PrecMin	1051	51	38	46	45	41	34	36	39	72	69	65	75	(17)
(18)		PrecStd	168	28	32	30	32	26	29	42	59	45	40	49	28	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.0	1.7	2.7	7.1	13.7	14.7	16.7	20.9	18.1	15.6	9.0	6.4	(19)
(20)			MCWB	1.3	0.8	1.2	3.6	7.3	10.8	14.1	18.3	14.9	11.9	7.5	5.2	(20)
(21)		2%	DB	1.3	0.7	1.2	5.0	9.8	10.9	14.3	17.9	16.1	12.3	7.7	4.6	(21)
(22)			MCWB	0.7	0.0	0.5	2.7	6.6	8.6	13.1	16.1	14.2	9.7	6.9	3.7	(22)
(23)		5%	DB	0.7	0.0	0.4	3.8	7.6	9.2	12.7	16.4	14.3	10.8	6.6	3.2	(23)
(24)			MCWB	0.1	-0.9	-0.2	2.4	5.2	7.7	11.6	15.2	12.2	9.0	5.9	2.7	(24)
(25)		10%	DB	-0.1	-0.8	-0.2	2.8	5.9	8.0	11.2	14.9	13.0	9.8	5.6	2.0	(25)
(26)			MCWB	-0.8	-1.6	-0.8	1.8	4.6	7.1	10.5	14.2	11.8	8.5	4.7	1.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.6	1.5	2.0	4.7	8.7	11.3	15.2	18.9	16.2	12.6	8.1	5.7	(27)
(28)			MCD B	1.7	1.7	2.3	5.6	12.2	12.8	16.0	20.1	16.8	14.8	8.4	5.7	(28)
(29)		2%	WB	0.8	0.4	0.7	3.6	6.8	9.1	13.4	16.8	14.3	10.6	7.2	3.9	(29)
(30)			MCD B	1.0	0.5	0.9	4.3	8.8	10.3	14.1	17.5	15.6	11.5	7.6	4.2	(30)
(31)		5%	WB	0.2	-0.6	0.0	2.7	5.6	8.0	11.9	15.3	12.9	9.5	6.1	2.7	(31)
(32)			MCD B	0.5	-0.2	0.3	3.4	7.0	8.7	12.5	15.8	13.6	10.2	6.5	3.3	(32)
(33)		10%	WB	-0.7	-1.6	-0.6	2.0	4.7	7.2	10.6	14.2	12.0	8.8	5.0	1.1	(33)
(34)			MCD B	-0.2	-0.9	-0.3	2.6	5.8	7.9	11.1	14.8	12.7	9.6	5.4	1.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	3.1	3.1	3.1	3.2	4.2	4.1	4.2	5.0	4.5	4.1	3.8	3.4	(35)
(36)			MCDBR	3.5	3.7	4.0	5.5	8.1	6.9	7.1	7.7	7.3	5.7	5.5	5.1	(36)
(37)			MCWBR	3.6	4.1	4.1	4.1	5.4	5.2	5.7	6.3	5.5	4.5	5.3	5.6	(37)
(38)		5% WB	MCD BR	3.5	4.0	3.7	4.9	6.8	5.8	6.9	6.9	6.4	5.2	5.4	4.9	(38)
(39)			MCWBR	3.7	4.4	4.0	3.9	5.1	4.9	5.9	6.2	5.4	4.5	5.5	5.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.354	0.375	0.438	0.488	0.495	0.482	0.458	0.437	0.392	0.399	0.392	0.366	(40)	
(41)		taud	2.128	2.088	1.961	1.895	1.961	2.083	2.231	2.305	2.383	2.189	2.106	2.077	(41)	
(42)		Ebn at Noon	698	768	768	768	781	794	809	812	820	735	642	623	(42)	
(43)		Edh at Noon	94	119	157	183	177	158	135	121	102	107	96	89	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.12	1.84	3.01	3.87	4.19	4.29	4.22	3.66	3.35	2.44	1.34	0.95	(44)	
(45)		RadStd	0.07	0.15	0.27	0.41	0.45	0.35	0.40	0.27	0.27	0.17	0.13	0.05	(45)	

Historical Trends

	DBAvg	Heating		Cooling		Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page